

- 1. (a)**  
Given two relation show that the  $\omega$  and  $\omega^2$  are the roots of  
$$\frac{1}{a+x} + \frac{1}{b+x} + \frac{1}{c+x} = \frac{2}{x}$$
which is cubic in  $x$  and we have to prove that 1 is roots of it  
$$x \Sigma (b+x)(c+x) = 2(a+x)(b+x)(c+x)$$
$$\Rightarrow x[\Sigma bc + 2(\Sigma a)x + 3x^2] = 2[x^3 + x^2 \Sigma a + x \Sigma ab + abc]$$
$$\Rightarrow 3x^3 + 2x^2 \Sigma a + x \Sigma bc = 2x^3 + 2x^2 \Sigma a + 2x \Sigma ab + 2abc$$
$$\Rightarrow x^3 + 0 \cdot x^2 - x \Sigma ab - 2abc = 0$$
It is cubic in  $x$  whose roots are  $\omega, \omega^2$ , and  
Let third root is  $\alpha$   
Now  $\omega + \omega^2 + \alpha = 0$   
 $\Rightarrow \alpha = 1$  ( $\because \omega + \omega^2 = -1$ )  
$$\Rightarrow \sum \frac{1}{a+1} = \frac{2}{1} = 2$$
- 2. (b)**  
 $(x_1 x_2 x_3 \dots \infty)^2 (z_1 z_2 z_3 \dots \infty)^4$   
$$\left[ \left( \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) \left( \cos \frac{\pi}{2^2} + i \sin \frac{\pi}{2^2} \right) \left( \cos \frac{\pi}{2^3} + i \sin \frac{\pi}{2^3} \right) \dots \infty \right]^2$$
$$\left[ \left( \cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \left( \cos \frac{\pi}{3^2} + i \sin \frac{\pi}{3^2} \right) \dots \infty \right]^4$$
$$= \left[ \cos \left( \frac{\pi}{3} + \frac{\pi}{3^2} + \frac{\pi}{3^3} + \dots \right) + i \sin \left( \frac{\pi}{3} + \frac{\pi}{3^2} + \frac{\pi}{3^3} + \dots \right) \right]^4$$
$$= \left[ \cos \left( \frac{\pi/2}{1-\frac{1}{2}} \right) + i \sin \left( \frac{\pi/2}{1-\frac{1}{2}} \right) \right]^2 \left[ \cos \left( \frac{\pi/3}{1-\frac{1}{3}} \right) + i \sin \left( \frac{\pi/3}{1-\frac{1}{3}} \right) \right]^4$$
$$(\cos \pi + i \sin \pi)^2 (\cos \pi/2 + i \sin \pi/2)^4 = (-1)^2 (i)^4 = 1$$
- 3. (c)**  
 $(Z-1) \cdot (\bar{Z}-1) = (Z-1) \cdot (\overline{Z-1}) = |Z-1|^2$   
( $\because Z\bar{Z} = |Z|^2$ )
- 4. (a)**  
 $Y\bar{Z} = a^2 + b^2 - ab \quad \therefore \phi \circ \Xi \Psi \bar{Z} = a^3 + b^3 X$   
&  $X+Y+\bar{Z} = 0 X$   
&  $X^2 + Y^2 + \bar{Z}^2 = a^2 + b^2 + 2ab + a^2\omega^2 + b^2\omega$   
 $+ 2ab + a^2\omega^4 + b^2\omega^2 + 2ab = 6ab X$
- 5. (b)**  
 $z = \omega, \omega^2$  (roots);  $z^{2003} = \omega^{2003} = \omega^2$   
$$z^{2003} + \frac{1}{2^{2003}} = \omega^2 + \frac{1}{\omega^2} = \omega^2 + \omega = -1$$
- 6. (c)**  
 $|z_1 + z_2| = |z_1| + |z_2|$   
 $|z_1|^2 + |z_2|^2 + 2|z_1||z_2|\cos(\theta_1 - \theta_2) = |z_1|^2 + |z_2|^2 + 2|z_1||z_2|$   
 $\Rightarrow \cos(\theta_1 - \theta_2) = 1 \Rightarrow \theta_1 - \theta_2 = 0$   
 $\arg(z_1) - \arg(z_2) = 0$
- 7. (c)**  
$$\boxed{z^2 = \bar{z}} \Rightarrow (\bar{z})^2 = \bar{\bar{z}} \Rightarrow (z^2)^2 = z \Rightarrow \boxed{z^4 = z}$$
  
 $z = 0$  or  $z^3 = 1 \Rightarrow \boxed{z = 0, 1, \omega, \omega^2}$   
Hence four solutions.
- 8. (a)**  
 $x_1 x_2 x_3 \dots \infty = c$  is  $\frac{\pi}{2} \times c$  is  $\frac{\pi}{2^2} \times c$  is  $\frac{\pi}{2^3} \dots \infty = c$  i  
$$\left( \frac{\pi}{2} + \frac{\pi}{2^2} + \frac{\pi}{2^3} \dots \infty \right)$$
$$\frac{\pi/2}{1-\frac{1}{2}} = i\pi = \cos \pi + i \sin \pi = -1$$
- 9. (c)**  
$$\frac{1+i\cos\theta}{1-2i\cos\theta} = \frac{(1+i\cos\theta)(1+2i\cos\theta)}{(1-2i\cos\theta)(1+2i\cos\theta)}$$
Given that  
$$= \left[ \frac{(1-2\cos^2\theta)}{(1+4\cos^2\theta)} \right] + i \left[ \frac{3\cos\theta}{1+4\cos^2\theta} \right]$$
Since  $\text{Im}(z) = 0$ , then  $3\cos\theta = 0 \Rightarrow \theta = n\pi \pm \pi/2$ .
- 10. (c)**  
Given equation  $(x^4 + 2xi) - (3x^2 + yi) = (3-5i) + (1+2yi)$   
 $\Rightarrow (x^4 - 3x^2) + i(2x - 3y) = 4 - 5i$   
Equating real and imaginary parts, we get  
 $x^4 - 3x^2 = 4$  .....(i) and  $2x - 3y = -5$  .....(ii)  
Form (i) and (ii), we get  $x = \pm 2$  and  $y = 3, \frac{1}{3}$ .  
**Trick:** Put  $x = 2, y = 3$  and then  $x = -2, y = \frac{1}{3}$ , we see that they both satisfy the given equation.
- 11. (b)**  
Let  $z = x + iy$ ,  $z + iz = (x-y) + i(x+y)$  and  $iz = -y + ix$   
If A denotes the area of the triangle formed by  $z, z + iz$  and  $iz$ , then  
$$A = \frac{1}{2} \begin{vmatrix} x & y & 1 \\ x-y & x+y & 1 \\ -y & x & 1 \end{vmatrix}$$
Applying transformation  $R_2 \rightarrow R_2 - R_1 - R_3$   
$$A = \frac{1}{2} \begin{vmatrix} x & y & 1 \\ 0 & 0 & -1 \\ -y & x & 0 \end{vmatrix} = \frac{1}{2} (x^2 + y^2) = \frac{1}{2} |z|^2$$
We get
- 12. (d)**  
 $|z-4| < |z-2| \Rightarrow |z-4|^2 < |z-2|^2 \Rightarrow (z-4)(\bar{z}-4)$   
 $< (z-2)(\bar{z}-2) \Rightarrow z\bar{z} - 4(z+\bar{z}) + 16 < z\bar{z} - 2(z+\bar{z}) + 4$   
 $\Rightarrow 2(z+\bar{z}) > 12 \Rightarrow 4 \text{Re}(z) > 12 \Rightarrow \text{Re}(z) > 3$
- 13. (c)**  
 $|z-2| = \min \{ |z-1|, |z-5| \}$   
i.e.,  $|z-2| = |z-1|$ , where  $|z-1| < |z-5|$

$$\Rightarrow \operatorname{Re}(z) = \frac{3}{2} \quad \text{which satisfy } |z - 1| < |z - 5|.$$

Also,  $|z - 2| = |z - 5|$ , where  $|z - 5| < |z - 1|$ .

$$\Rightarrow \operatorname{Re}(z) = \frac{7}{2} \quad \text{which satisfy } |z - 5| < |z - 1|.$$

14. (d)

$$(1+i)^{n_1} + (1+i^3)^{n_1} + (1+i^5)^{n_2} + (1+i^7)^{n_2} =$$

$$(1+i)^{n_1} + (1-i)^{n_1} + (1+i)^{n_2} + (1-i)^{n_2} = \text{real number for all}$$

$n_1$  and  $n_2$

15. (d)

On putting  $z = x + iy$  the equation is same as

$$|x^2 - y^2 + 2ixy - 1| = x^2 + y^2 + 1$$

$$\Rightarrow (x^2 - y^2 - 1)^2 + 4x^2y^2 = (x^2 + y^2 + 1)^2$$

$$\Rightarrow x = 0$$

$\Rightarrow z$  lies on imaginary axis, so (A), (B), (C) are ruled out.

16. (c)

Let the vertices of the square ABCD are 1, -1, i and -i in the Argand Diagram.

Let P be represented by z, then  $\angle PAB + \angle PBC + \angle PCD + \angle PDA$

$$= \arg \left( \frac{z^4 - 1}{4} \right) = \arg(z^4 - 1).$$

Since  $z^4$  also lies inside the square

$$\Rightarrow \frac{5\pi}{4} \geq \arg(z^4 - 1) \geq \frac{3\pi}{4}$$

17. (d)

If  $Z_1 = r_1 e^{i\theta}$ ,  $Z_2 = r_2 e^{i(\theta+\alpha)}$ ,  $ar_1 = br_2$  and

$$\frac{aZ_1 - bZ_2}{aZ_1 + bZ_2} = \frac{e^{i\theta} - e^{i(\theta+\alpha)}}{e^{i\theta} + e^{i(\theta+\alpha)}} = \frac{1 - e^{i\alpha}}{1 + e^{i\alpha}}$$

$$= \frac{e^{-i\alpha/2} - e^{i\alpha/2}}{e^{-i\alpha/2} + e^{i\alpha/2}}$$

$$= \frac{-2i \sin \frac{\alpha}{2}}{2 \cos \frac{\alpha}{2}} = -i \tan \frac{\alpha}{2}$$

implying that the point is on the imaginary axis.

18. (d)

$|z + a| \leq a \Rightarrow z$  lies on or inside circle with centre  $(-a, 0)$  & radius  $a$

$$\text{So } |z_1| + |z_2| + |z_3| \leq 14$$

19. (b)

$$\text{from } |z - 4 + 3i| \leq 1$$

$$\Rightarrow |z|_{\max} = 6, |z|_{\min} = 4$$

$$\text{So } \alpha = 4, \beta = 6$$

$$y = \frac{x^4 + x^2 + 4}{x} = x^3 + x + \frac{4}{x}$$

For y to be least  $x = 1$

$$\Rightarrow y = 6 \Rightarrow y = \beta \Rightarrow k = \beta$$

20. (b)

$$\therefore |z_1 - z_2| \geq ||z_1| - |z_2||$$

$$\therefore |z_2 - 3 - 4i| \geq ||z_2| - |3 + 4i|| \Rightarrow 5 \geq ||z_2| - 5|$$

$$\Rightarrow -5 \leq |z_2| - 5 \leq 5$$

$$\Rightarrow |z_2| \leq 10 \Rightarrow -|z_2| \geq -10$$

$$\therefore |z_1| - |z_2| \geq 12 - 10 = 2$$

But  $|z_1 - z_2| \geq |z_1| - |z_2| \geq 2 \Rightarrow$  Minimum value of  $|z_1 - z_2|$  is 2.